

Visualizing Lattice-based Post-Quantum Cryptographic Algorithms

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Abstract. The abstract should briefly summarize the contents of the paper in 150–250 words.

Keywords: Cryptography · Post-Quantum · Another keyword.

1 Introduction

Cryptography is the study of techniques and protocols relating to securing communications (Rivest, 1990). Typically, this consists of constructing and analyzing protocols that prevent outside parties from reading private messages.

Prior to the modern era, most of cryptography was encryption, which is a process that transforms some form of information (plaintext) into an alternative form that only certain parties can decode (ciphertext). What is now known as classical cryptography consisted of using ciphers that could be practically computed and solved by hand. However, after devices such as the Enigma machine and other computational devices were developed, these ciphers became trivially easy to break.

Now, history repeats itself with the development of quantum computers as they threaten to render multiple modern cryptographic algorithms useless. Quantum algorithms such as the ones proposed by Peter Shor (Shor, 1997) are theorized to be able to break protocols like RSA given a strong enough quantum computer. Because of this, cryptology experts around the world have looked into new protocols that might have the potential to defend against them.

NIST (National Institute of Standards and Technology) is one of the organizations at the forefront of the effort. In 2016, they put out a call for proposals that might be used in post-quantum cryptography. Eight years later, they released three standards based on those submissions, two of which are lattice-based; ML-KEM and ML-DSA, based on the Kyber and Dilithium algorithms respectively. ML-KEM is used for key encapsulation, while ML-DSA is used for digital signatures.

Despite their importance, there are a limited number of resources available for learning about them. In some part this is because they are relatively new

standards. However, it can also be attributed to their complexity and nature as lattice-based. Aside from NIST’s released papers (NIST, 2024), attempts have been made to visualize the scheme and other lattice-based algorithms, including the following: - pqc-vizz (ML-KEM, ML-DSA, SLH-DSA) - CrypTool2 (cryptography module) However, they tend to skip parts of the step-by-step process, not show the background calculations, as well as simplify the lattice too much (Rodríguez et al., 2024; Kalava et al., 2025).

Because of the above, this work aims to provide a visualization that allows

1.1 A Subsection Sample

Please note that the first paragraph of a section or subsection is not indented. The first paragraph that follows a table, figure, equation etc. does not need an indent, either.

Subsequent paragraphs, however, are indented.

1.2 Related Works

Sample Heading (Third Level) Only two levels of headings should be numbered. Lower level headings remain unnumbered; they are formatted as run-in headings.

Sample Heading (Fourth Level) The contribution should contain no more than four levels of headings. Table 1 gives a summary of all heading levels.

Table 1. Table captions should be placed above the tables.

Heading level	Example	Font size and style
Title (centered)	Lecture Notes	14 point, bold
1st-level heading	1 Introduction	12 point, bold
2nd-level heading	2.1 Printing Area	10 point, bold
3rd-level heading	Run-in Heading in Bold. Text follows	10 point, bold
4th-level heading	<i>Lowest Level Heading.</i> Text follows	10 point, italic

Displayed equations are centered and set on a separate line.

$$x + y = z \tag{1}$$

Please try to avoid rasterized images for line-art diagrams and schemas. Whenever possible, use vector graphics instead (see Fig. 1).

Theorem 1. *This is a sample theorem. The run-in heading is set in bold, while the following text appears in italics. Definitions, lemmas, propositions, and corollaries are styled the same way.*

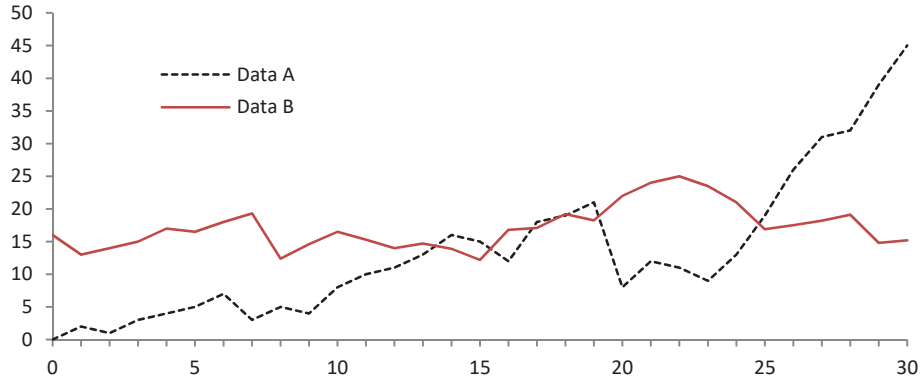


Fig. 1. A figure caption is always placed below the illustration. Please note that short captions are centered, while long ones are justified by the macro package automatically.

Proof. Proofs, examples, and remarks have the initial word in italics, while the following text appears in normal font.

For citations of references, we prefer the use of square brackets and consecutive numbers. Citations using labels or the author/year convention are also acceptable. The following bibliography provides a sample reference list with entries for journal articles [1], an LNCS chapter [2], a book [3], proceedings without editors [4], and a homepage [5]. Multiple citations are grouped [1–3], [1, 3–5].

Acknowledgments. A bold run-in heading in small font size at the end of the paper is used for general acknowledgments, for example: This study was funded by X (grant number Y).

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2 Introduction

2.1 A Subsection Sample

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References

1. Author, F.: Article title. *Journal* **2**(5), 99–110 (2016)
2. Author, F., Author, S.: Title of a proceedings paper. In: Editor, F., Editor, S. (eds.) *CONFERENCE 2016, LNCS*, vol. 9999, pp. 1–13. Springer, Heidelberg (2016). <https://doi.org/10.10007/1234567890>
3. Author, F., Author, S., Author, T.: Book title. 2nd edn. Publisher, Location (1999)
4. Author, A.-B.: Contribution title. In: *9th International Proceedings on Proceedings*, pp. 1–2. Publisher, Location (2010)
5. LNCS Homepage, <http://www.springer.com/lncs>, last accessed 2023/10/25